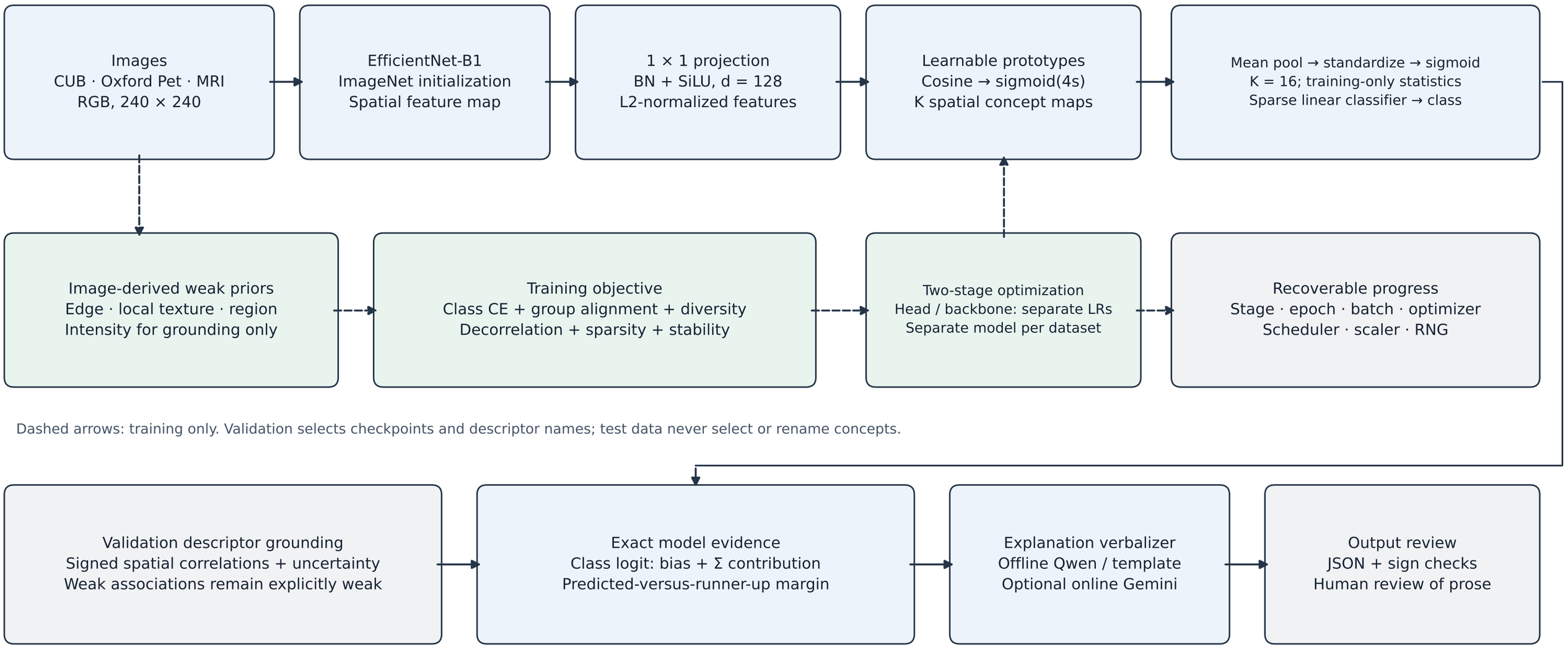
RG-CBM: descriptor-grounded concept bottleneck for visual classification

Shared architecture; separate trained model per dataset. No manually annotated concepts or masks in the training objective.



Dashed arrows: training only. Validation selects checkpoints and descriptor names; test data never select or rename concepts.

Independent evaluation: accuracy / macro-F1 / calibration · concept ablation vs random · stability · held-out alignment · CUB boxes / Pet trimaps

Descriptor association is not a part annotation, tumor mask, clinical finding, or causal claim about the image. LLM output never feeds the classifier.